



Set By: BRA

Teachers:

*BRA
MAX
HAY

KNOX GRAMMAR SCHOOL
MATHEMATICS DEPARTMENT

2006
HALF YEARLY EXAMINATION

Mathematics Extension 2

(Year 12)

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen
- Board-approved calculators may be used
- All necessary working should be shown in every question
- A table of standard integrals is provided at the back of this paper

Total marks (120)

- Attempt Questions 1–8
- All questions are of equal value
- Use a **SEPARATE** writing booklet for each question
- Write your Board of Studies (BOS) Number and Teacher's Initials on the front cover of each of your writing booklets

BOS NUMBER: _____

TEACHER'S INITIALS: _____

BLANK PAGE

Total marks (120)
Attempt questions 1 – 8
All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (15 marks) Use a SEPARATE writing booklet **Marks**

(a) Find z in the form $a + bi$ if $z = \frac{2-i}{2+i}$. **2**

(b) If $z = -1 + i\sqrt{3}$ determine:

(i) $\bar{z}$ **1**

(ii) $|z|$ **1**

(iii) z^4 (In Cartesian Form) **3**

(iv) $\text{Arg}(z^4)$ (Exact Value In Radians) **1**

(c) Draw a neat, accurate sketch of the region representing the complex number z where z satisfies the inequalities $1 \leq |z-1| \leq 2$ and $\frac{\pi}{4} \leq \arg(z) \leq \frac{\pi}{2}$. **3**

(d) The complex number z satisfies the equation $\arg(z+2) = \frac{\pi}{3}$.

(i) Sketch the points z on an Argand diagram for which $\arg(z+2) = \frac{\pi}{3}$. **2**

(ii) What is the minimum value of $|z|$? **2**

Question 2 (15 marks)

Use a SEPARATE writing booklet

Marks

(a) Prove by induction that $\sin x + \sin 2x + \sin 3x + \cdots + \sin nx = \frac{\sin\left(\frac{nx}{2}\right)\sin\left(\frac{n+1}{2}x\right)}{\sin\left(\frac{x}{2}\right)}$ 4

for all counting numbers n .

(Hint: You may assume that $2 \cos A \sin B = \sin(A+B) - \sin(A-B)$.)

(b) (i) Given $z = \cos \theta + i \sin \theta$, simplify $z^n - z^{-n}$ where n is a counting number. 2

(ii) By expanding $(z - z^{-1})^5$ find the values of the constants a , b and c so that: 4

$$\sin^5 \theta = a \sin 5\theta + b \sin 3\theta + c \sin \theta.$$

(Hint: Use the coefficients 1, 5, 10, 10, 5, 1 in the expansion of $(z - z^{-1})^5$.)

(c) ω is a complex cube root of unity.

(i) Prove that the other complex root is ω^2 . 2

(ii) Prove that $1 + \omega + \omega^2 = 0$. 1

(iii) Simplify $(1 - 3\omega + \omega^2)(1 + \omega - 8\omega^2)$. 2

Question 3 (15 marks)

Use a SEPARATE writing booklet

Marks

(a) Factorise $x^3 - 8$ over the field of

(i) real numbers

1

(ii) complex numbers

2

(b) $(x - 2)$ is a factor of $P(x) = x^3 + ax + b$. When $P(x)$ is divided by $(x + 3)$ the remainder is -10 .

(i) Find the values of a and b .

3

(ii) Using the values for a and b obtained in (i), what is the remainder when $P(x)$ is divided by $(x - 2)(x + 3)$?

2

(c) Given that $2 + 3i$ is a root of the equation $x^4 - 4x^3 + 4x^2 + 36x - 117 = 0$, find the other roots.

4

(d) If α, β and γ are the roots of $2x^3 - 3x^2 - 3x + 2 = 0$, form the cubic equation whose roots are α^2, β^2 and γ^2 .

3

Question 4 (15marks) Use a SEPARATE writing booklet

Marks

- (a) Sketch the graph of $y = \ln x$. Use your graph to sketch on separate number planes the graphs of:
- (i) $y = |\ln x|$
- (ii) $y = \ln|x|$
- (iii) $|y| = \ln x$
- (iv) $y = \ln(-x)$
- (b) (i) If $f(x) = \tan^{-1} x + \tan^{-1}\left(\frac{1}{x}\right)$ for $x > 0$, show that $f'(x) = 0$.
- (ii) Hence, or otherwise, sketch the graph of $f(x)$.
- (c) On a number plane, sketch the region given by $y \leq |x^2 - 1|$.
- (d) Without using calculus, draw a neat sketch of $y = \frac{1}{(x-1)(x+2)}$ showing all important features.

- (a)
- (i) Show that if the polynomial equation $P(x) = 0$ has a root α of multiplicity m then the equation $P'(x) = 0$ has a root of multiplicity $m - 1$. **3**
- (ii) Find the roots of the equation $x^4 + 5x^3 - 21x^2 + 23x - 8 = 0$ given that it has a root of multiplicity 3. **3**
- (b) Find the values of p and q such that $x^4 + 6x^3 + 13x^2 + px + q$ will be a perfect square. **3**
- (c) If α , β and γ are the roots of the cubic equation $x^3 - 5x^2 + 2x - 1 = 0$, write down the values of :
- (i) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ **2**
- (ii) $\alpha^2 + \beta^2 + \gamma^2$ **2**
- (iii) $\frac{\alpha\beta}{\gamma} + \frac{\alpha\gamma}{\beta} + \frac{\beta\gamma}{\alpha}$ **2**

- (a) Sketch the following graphs on separate number planes, clearly showing all features.

(i) $y = \tan^{-1}|x|$ 2

(ii) $y = \frac{\pi}{2} + \sin^{-1}(x-1)$ 2

(iii) $\sin(x+y) = 0$ 2

$$\begin{aligned}\sin^{-1}(\theta) &= x+y \\ \sin^{-1}(\theta) - x &= y\end{aligned}$$

- (b) You are given that the graphs for the following equations each have just one line of symmetry. State in each case the equation of the line of symmetry, giving reasons for your answers.

(i) $x^4 = y^3$ 2

(ii) $x^3 + y^3 = 1$ 2

- (c) (i) Find the equation of the locus of w if $w = \frac{z-2}{z-1}$ and 3

z describes the unit circle centred at the origin.

- (ii) Hence, sketch the locus of w on an Argand diagram. 2

Question 7 (15 marks)

Use a SEPARATE writing booklet

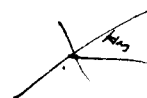
Marks

- (a) State the domain and range of $y = \sin^{-1}\left(\frac{x}{2}\right)$. 2

- (b) (i) Write $\sin \theta + \sqrt{3} \cos \theta$ in the form $A \cos(\theta - \alpha)$ where $A > 0$ and α is acute. 2

- (ii) Hence, solve $\sin \theta + \sqrt{3} \cos \theta = 1$ for $0 \leq \theta \leq 2\pi$. 2

- (c) Find, in terms of π , the value of:



- (i) $\int_0^2 \frac{1}{4+x^2} dx$ 2

- (ii) $\int_0^{\frac{1}{6}} \left(\frac{1}{\sqrt{1-9x^2}} \right) dx$ 2

- (d) If n is a counting number, n factorial, which is written as $n!$ is defined by:

$$n! = n(n-1)(n-2) \cdots 3 \cdot 2 \cdot 1$$

For example, $4! = 4 \cdot 3 \cdot 2 \cdot 1$ or 24.

- (i) Find the smallest counting number A for which $n! \geq 3^n$. 1

- (ii) Hence, prove by mathematical induction that $n! \geq 3^n$ for $n \geq A$. 4

- (a) Find the general solution to $2\sin^2 \theta + \cos \theta = 1$. 4
- (b) Differentiate $\cos^4 \theta$ and use the result to evaluate $\int_0^{\pi/4} (\cos^3 \theta \sin \theta) d\theta$ 3
- (c) Find $\int \frac{1 - \cos 2\theta}{1 + \cos 2\theta} d\theta$ 2
- (d) (i) Write down $\sin \theta, \cos \theta$ and $\tan \theta$ in terms of t where $t = \tan \frac{\theta}{2}$ 1
- (ii) Use these results to prove that $\sec \theta - \tan \theta = \frac{1-t}{1+t}$. 2
- (iii) Hence, find $\int (\sec \theta - \tan \theta) d\theta$ 3

End of Paper

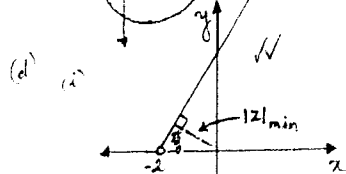
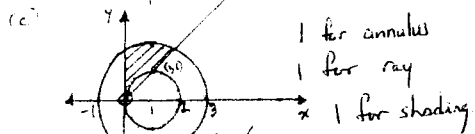
SOLUTIONS

(a) $\frac{2-i}{2+i} \cdot \frac{2-i}{2-i} = \frac{(2-i)^2}{2^2+1^2}$ or $\frac{3}{5} - \frac{4}{5}i$

(b) (i) $\bar{z} = -1 - \sqrt{3}i$
(ii) $|z| = \sqrt{(-1)^2 + (\sqrt{3})^2} = 2$

(iii) $z = 2 \cos \frac{2\pi}{3} + 2i \sin \frac{2\pi}{3} = 1 + \sqrt{3}i$
 $= 10 \cos \frac{2\pi}{3} + 8i \sin \frac{2\pi}{3} = -8 + 8\sqrt{3}i$

(iv) $\text{Arg}(z^4) = \frac{2\pi}{3}$



(ii) $\sin \frac{\pi}{3} = \frac{|z|_{\min}}{2}$
 $\Rightarrow |z|_{\min} = \sqrt{3}$

1. When $n=1$, LHS = $\sin x$, RHS = $\frac{\sin \frac{x}{2} \cdot \sin x}{\sin \frac{x}{2}}$ or $\sin x$ true for $n=1$.
If the statement is true for $n=k$, $k \in \mathbb{N}$, then:-
 $\sin x + \sin kx + \sin(k+1)x = \frac{\sin \frac{kx}{2} \cdot \sin(\frac{k+1}{2}x)}{\sin \frac{x}{2}} + \sin(k+1)x$

$$\frac{\sin \frac{kx}{2} \cdot \sin(\frac{k+1}{2}x)}{\sin \frac{x}{2}} + 2 \sin(\frac{k+1}{2}x) \cdot \cos(\frac{k+1}{2}x)$$

$$\frac{\sin(\frac{k+1}{2}x)}{\sin \frac{x}{2}} \left[\sin \frac{kx}{2} + 2 \cos(\frac{k+1}{2}x) \cdot \sin \frac{x}{2} \right]$$

$$\frac{\sin(\frac{k+1}{2}x)}{\sin \frac{x}{2}} \left[\sin \frac{kx}{2} + \sin(\frac{k+2}{2}x) - \sin \frac{kx}{2} \right]$$

$$\frac{\sin(\frac{k+1}{2}x)}{\sin \frac{x}{2}} \cdot \sin(\frac{k+2}{2}x)$$

ie, if true for $n=k \Rightarrow$ true for $n=k+1$. As it is true for $n=1$, then by induction it is true $\forall n \in \mathbb{N}$.

$z^n - z^{-n} = 2i \sin n\theta$ (Consequence of R.P.M.)
 $z^5 - z^{-5} = z^5 - 5z^4z^{-1} + 10z^3z^{-2} - 10z^2z^{-3} + 5zz^{-4} - z^{-5}$
 $(z^5 - z^{-5}) - 5(z^3 - z^{-3}) + 10(z - z^{-1})$

$(2i \sin \theta)^5 = 2i \sin 5\theta - 5(2i \sin 3\theta) + 10(2i \sin \theta)$
 $\Rightarrow \sin 5\theta = \frac{1}{16} \sin 5\theta - \frac{5}{16} \sin 3\theta + \frac{5}{8} \sin \theta$

$\therefore a = \frac{1}{16}, b = -\frac{5}{16} \text{ \& } c = \frac{5}{8}$

(c) (i) $\omega^3 = 1 \Rightarrow (\omega^3)^2 = 1^2$
 $\Rightarrow (\omega^2)^3 = 1$

ω is also a complex root of unity

(ii) From $z^3 - 1 = 0 \Rightarrow \sum \alpha = 0$
 $\Rightarrow 1 + \omega + \omega^2 = 0$

(iii) $(1 - 3\omega + \omega^2)(1 + \omega - 9\omega^2)$
 $= (1 + \omega + \omega^2 - 4\omega)(1 + \omega + \omega^2 - 9\omega^2)$
 $= (-4\omega)(-9\omega^2)$ using (ii)
 $= 36\omega^3$ or 36

3. (b) $P(2) = 0$ (Fash-thin) & $P(-3) = -10$ (hem-thin)

(i) $\Rightarrow \begin{cases} 2a + b = -8 & \text{(i)} \\ -3a + b = 17 & \text{(ii)} \end{cases}$

$a = -5 \text{ \& } b = 2$

(ii) By div. transformation:
 $z^3 - 5x + 2 \equiv (z-2)(x+3)\delta(x) + px + q$
 $p, q \in \mathbb{R}$

$\Rightarrow \begin{cases} 2p + q = 0 & (\text{sub. } x=2) \\ -3p + q = -10 & (\text{sub. } x=-3) \end{cases}$

$\Rightarrow p = 2; q = -4$

Remainder = $2x - 4$

(i) $(x-2)(x^2 + 2x + 4)$

(ii) $(x-2)[(x+1)^2 - (\sqrt{3}i)^2]$ completing square

$= (x-2)(x+1+i\sqrt{3})(x+1-i\sqrt{3})$

(c) Co-effts real $\Rightarrow 2-3i$ is also a root.
 $\therefore (x-(2+3i))(x-(2-3i))$ are linear factors.

$\Rightarrow x^2 - 4x + 13$ is a quad. factor.

$\therefore x^4 - 4x^3 + 4x^2 + 36x - 117 = (x^2 - 4x + 13)(x^2 - 9)$
by inspection or equivalent
 $\Rightarrow$ other roots are $2-3i, \pm 3$

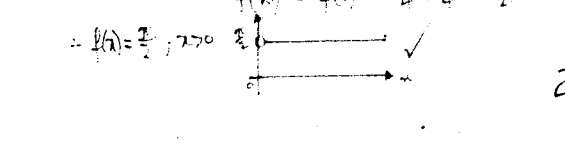
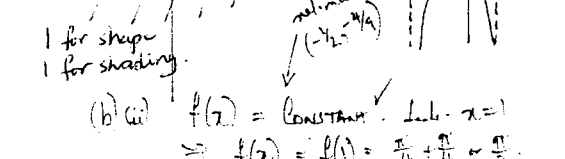
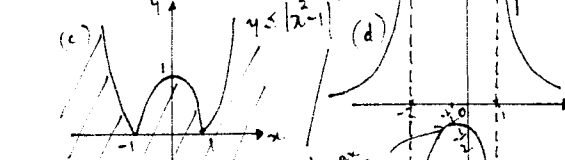
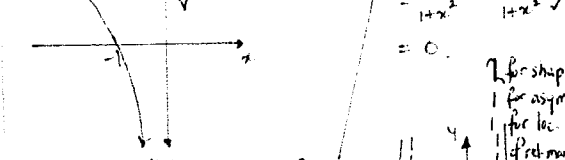
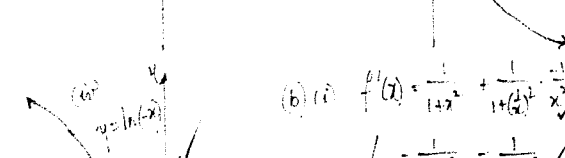
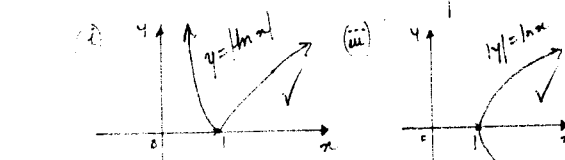
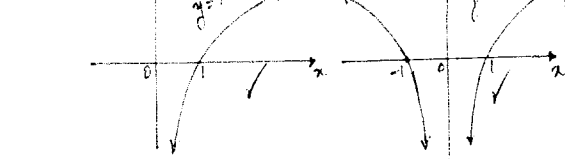
83. (cont.) (d) Let $z = iy \Rightarrow 2y^2y - 3y - 3iy + 2 = 0$
and $\alpha^3, \beta^3, \gamma^3$ satisfy this eqn. iff
 α, β, γ satisfy $2\alpha^3 - 3\alpha - 3\alpha i + 2 = 0$

$\Rightarrow y(2y-3) = 3y-2$

$\Rightarrow 4(2y-3)^2 = (3y-2)^2$

$\Rightarrow 20y^3 - 12y^2 + 9y = 9y^2 - 12y + 4$

$\Rightarrow 2y^3 - 21y^2 + 21y - 4 = 0$



5. (a) $P(x) = (x-2)^m \delta(x) \cdot 1; \delta(x) \neq 0$
 $\Rightarrow P'(x) = (x-2)^m \delta'(x) + m(x-2)^{m-1} \cdot \delta(x)$
 $= (x-2)^{m-1} [(x-2) \cdot \delta'(x) + m \delta(x)]$
 $= (x-2)^{m-1} \cdot \delta_1(x) \text{ \& } \delta_1(x) \neq 0$

(ii) Let $P(x) = x^4 + 5x^3 - 21x^2 + 23x - 8$
 $\Rightarrow P'(x) = 4x^3 + 15x^2 - 42x + 23$
 $\Rightarrow P''(x) = 12x^2 + 30x - 42$

$\therefore P''(x) = 0$ when $6(2x^2 + 5x - 7) = 0$
 $\Rightarrow 6(2x+7)(x-1) = 0$
 $\Rightarrow x = -\frac{7}{2} \text{ or } 1$

As $P(1) = 0 = P'(1) \Rightarrow 3$ fold root is $x=1$.
 $\therefore P(x) = (x-1)^3(x+8)$ by inspection.
 $\Rightarrow$ roots are $-8, 1$ (3 fold).

(b) $(x^2 + ax + b)^2 = x^4 + 6x^3 + 13x^2 + px + q$
for some $a, b \in \mathbb{R}$.

Equating co-effts of $x^3 \Rightarrow 2a = 6$
 $\Rightarrow a = 3$

Equating co-effts of $x^2 \Rightarrow a^2 + 2b = 13$
 $\Rightarrow b = 2$

$\therefore (x^2 + 3x + 2)^2 = x^4 + 6x^3 + 13x^2 + px + q$

Equating co-effts of $x \Rightarrow p = 12$

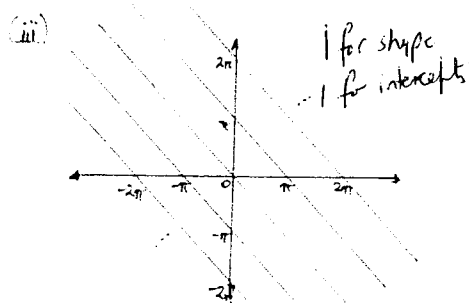
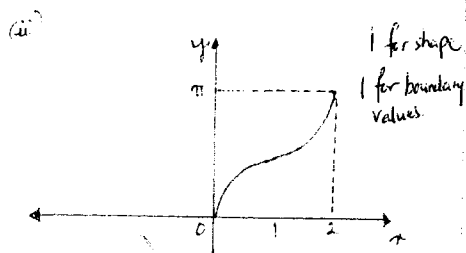
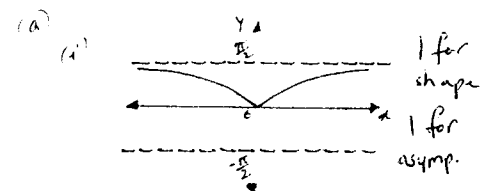
Equating constant terms $\Rightarrow q = 4$

(c) (i) $\sum \frac{1}{\alpha} = \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$
 $= \frac{\alpha\beta + \alpha\gamma + \beta\gamma}{\alpha\beta\gamma}$

$= \frac{2}{-(-1)} \text{ or } \underline{\underline{2}}$

(ii) $\sum \alpha^2 = \alpha^2 + \beta^2 + \gamma^2$
 $= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$
 $= (-(-5))^2 - 2 \cdot 2 \text{ or } \underline{\underline{21}}$

(iii) $\sum \frac{\gamma}{\alpha\beta} = \frac{\alpha}{\beta\gamma} + \frac{\beta}{\alpha\gamma} + \frac{\gamma}{\alpha\beta}$
 $= \frac{\alpha + \beta + \gamma}{\alpha\beta\gamma} = \frac{2}{-(-1)} \text{ or } \underline{\underline{2}}$



(i) $x = 0$ (y axis) as $(-x)^n = x^n$ ✓

(ii) $y = x$ ✓ as interchanging x & y introduces the same eq'n ✓

(iii) $|z| = 1$ ✓

Let z be the subject: $wz - w = z - 2$

$$\Rightarrow z(w-1) = w-2$$

$$\Rightarrow z = \frac{w-2}{w-1}$$

$$\Rightarrow 1 = \left| \frac{w-2}{w-1} \right| \sqrt{\text{as } |z| = 1}$$

$$\Rightarrow |w-1| = |w-2| \quad \checkmark$$

w describes the $\perp$ bisector of the interval from $(1,0)$ to $(2,0)$.

i.e. w describes the vertical line $x = 1.5$.



7. (a) Domain $-2 \leq x \leq 2$; Range $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$.

(b) (i) $A = \sqrt{1^2 + (\pi)^2}$
 $= 2 \quad \checkmark$

(ii) $2\left(\cos \theta \cdot \frac{\sqrt{3}}{2} + \sin \theta \cdot \frac{1}{2}\right) = 2 \cos\left(\theta - \frac{\pi}{6}\right)$

$\Rightarrow \begin{cases} \cos \theta = \frac{\sqrt{3}}{2} \\ \sin \theta = \frac{1}{2} \end{cases} \Rightarrow \theta = \frac{\pi}{6} \quad \checkmark$

$\therefore \sin \theta + \sqrt{3} \cos \theta = 2 \cos\left(\theta - \frac{\pi}{6}\right)$

(ii) $\cos\left(\theta - \frac{\pi}{6}\right) = \frac{1}{2}$

$\Rightarrow \theta - \frac{\pi}{6} = \dots, -\frac{\pi}{3}, \frac{\pi}{3}, 2\pi - \frac{\pi}{3}, 2\pi + \frac{\pi}{3}, \dots$

$\Rightarrow \theta = \frac{\pi}{2}, \frac{11\pi}{6}$ only for $0 \leq \theta \leq 2\pi$.

(c) (i) $\left[\frac{1}{2} \tan^{-1}\left(\frac{x}{3}\right)\right]^2 = \frac{1}{2} (\tan^{-1}(1) - \tan^{-1}(0))$
 $= \frac{\pi}{8} \quad \checkmark$

(ii) $\frac{1}{3} \int_0^{\sqrt{3}} \frac{1}{\sqrt{\left(\frac{x}{3}\right)^2 + 1}} dx = \left[\frac{1}{3} \tan^{-1}\left(\frac{x}{3}\right)\right]_0^{\sqrt{3}}$
 $= \frac{1}{3} \cdot \tan^{-1}\left(\frac{\sqrt{3}}{3}\right) = \frac{1}{3} \cdot \frac{\pi}{6} = \frac{\pi}{18} \quad \checkmark$

(d) (i)

n	n!	3^n
1	1	3
2	2	9
3	6	27
4	24	81
5	120	243
6	720	729
7	5040	2187
8	40320	6561
9	362880	19683

1st occurrence
Hence, $A = 7$ ✓

(ii) From (i) statement is true for $n=7$ ✓

Assume $K! \geq 3^K$; K some integer ≥ 7 .

Then $(K+1)! = (K+1) \cdot (K)! \quad \checkmark$

$\geq (K+1) \cdot 3^K$ ✓ by induction hypothesis

$\geq 3 \cdot 3^K$ ✓ as K is at least 7

Hence, if it is true for $n=K$, then also true for $n=K+1$. As it is true for $n=7$, conclude by induction that it is true for all higher counting numbers.

8. (a) $2 \sin^2 \theta + \cos \theta = 1$

$\Rightarrow 2(1 - \cos^2 \theta) + \cos \theta = 1$

$\Rightarrow 2 \cos^2 \theta - \cos \theta - 1 = 0$

$\Rightarrow (2 \cos \theta + 1)(\cos \theta - 1) = 0 \quad \checkmark$

$\Rightarrow \cos \theta = -\frac{1}{2} \text{ or } 1$ ✓

$\Rightarrow \theta = 2k\pi \pm \cos^{-1}\left(-\frac{1}{2}\right)$ or

$2k\pi \pm \cos^{-1}(1) \quad \checkmark$

$\Rightarrow \theta = 2k\pi \pm \left(\pi - \frac{\pi}{3}\right)$ ✓

$2k\pi \pm 0$

$\Rightarrow \theta = 2k\pi \pm \frac{2\pi}{3}, 2k\pi; k \in \mathbb{Z}$

(b) $\frac{d}{d\theta} (\cos^4 \theta) = 4 \cos^3 \theta \cdot -\sin \theta$
 $= -4 \cos^3 \theta \cdot \sin \theta$ ✓ by the chain rule.

$\therefore \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos^3 \theta \cdot \sin \theta \cdot d\theta = -\frac{1}{4} \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} 4 \cos^3 \theta \cdot \sin \theta \cdot d\theta$

$= \left[-\frac{1}{4} \cos^4 \theta\right]_{\frac{\pi}{4}}^{\frac{\pi}{2}} \quad \checkmark$

$= -\frac{1}{4} \left[\cos^4 \frac{\pi}{2} - \cos^4 \frac{\pi}{4}\right]$

$= -\frac{1}{4} \left[\frac{1}{4} - 1\right]$

$= \frac{3}{16} \quad \checkmark$

(c) $\int \frac{2 \sin^2 \theta}{2 \cos^2 \theta} \cdot d\theta = \int \tan^2 \theta \cdot d\theta$

$= \int (\sec^2 \theta - 1) \cdot d\theta$

$= \tan \theta - \theta + C \quad \checkmark$

(d) (i) $\sin \theta = \frac{2t}{1+t^2}$; $\cos \theta = \frac{1-t^2}{1+t^2}$; $\tan \theta = \frac{2t}{1-t^2}$ ✓

(ii) $\sec \theta - \tan \theta = \frac{1}{\cos \theta} - \frac{\sin \theta}{\cos \theta}$
 $= \frac{1 - \sin \theta}{\cos \theta} \quad \checkmark$
 $= \frac{1 - \frac{2t}{1+t^2}}{\frac{1-t^2}{1+t^2}} \quad \text{by } \frac{1+t^2}{1+t^2}$
 $= \frac{1+t^2-2t}{1-t^2}$
 $= \frac{t^2-2t+1}{1-t^2}$
 $= \frac{(1-t)^2}{(1-t)(1+t)} \quad \checkmark$
 $= \frac{1-t}{1+t}$

(iii) $\int \frac{1 - \tan \frac{\theta}{2}}{1 + \tan \frac{\theta}{2}} d\theta \quad \checkmark$
 $= \int \frac{\tan \frac{\pi}{4} - \tan \frac{\theta}{2}}{1 + \tan \frac{\pi}{4} \tan \frac{\theta}{2}} d\theta \quad \checkmark$
 $= \int \tan\left(\frac{\pi}{4} - \frac{\theta}{2}\right) d\theta \quad \checkmark$
 $= 2 \ln \left| \cos\left(\frac{\pi}{4} - \frac{\theta}{2}\right) \right| + C$